Why edges

Pattern engines begin with polygons and their edges on named layers

The idea

- GOLDEN: 5 shapes, 3 layers, 20 edges (4 per shape), 3 shapes on M1

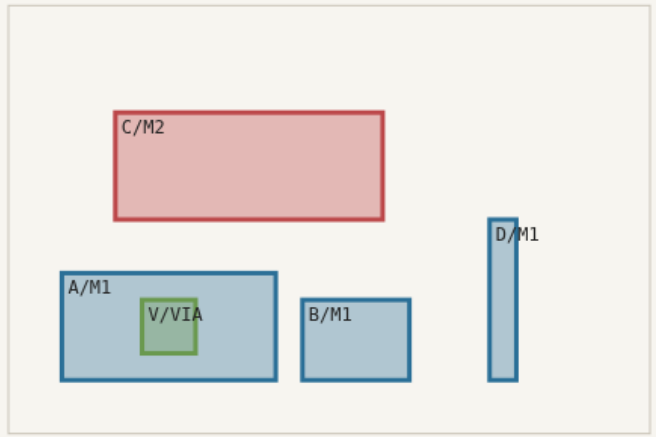
Shapes

EDGES AND POLYGONS

Shapes

Step 1 / 5 · shapes · module01-01-edges-polygons

shapes=5



Explanation

Five rectangles: A,B,D on M1; C on M2; V on VIA.

- layers

n=5

Shapes

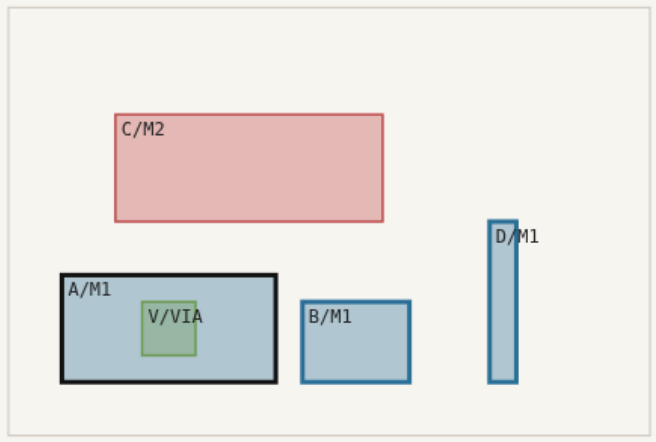
Layers

EDGES AND POLYGONS

Layers

Step 2 / 5 · layers · module01-01-edges-polygons

layers=3



Explanation

Layers M1, M2, VIA — GOLDEN layer_count=3.

- M1/M2/VIA

```
layers=3
```

Layers

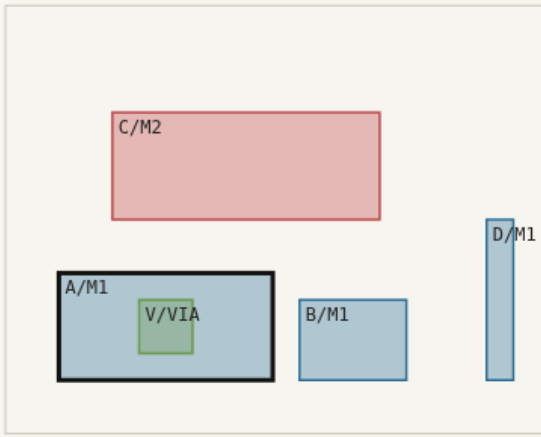
Edges

EDGES AND POLYGONS

Edges

Step 3 / 5 · edges · module01-01-edges-polygons

edges=20



Explanation

Each rect yields 4 edges → 20 total.

- 4×5

```
edges=20
```

Edges

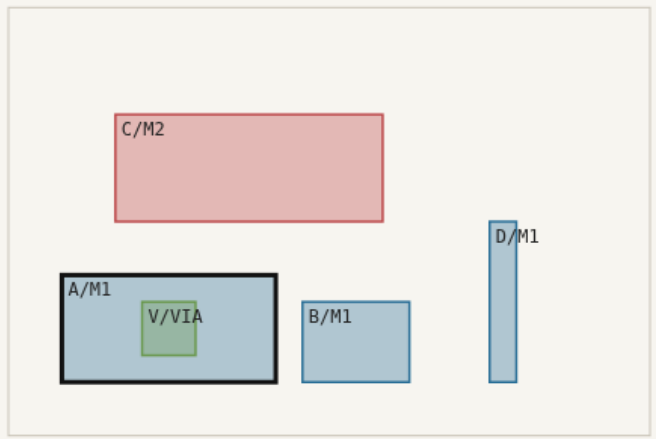
Select A

EDGES AND POLYGONS

Select A

Step 4 / 5 · a · module01-01-edges-polygons

A 8×4



Explanation

A is 8×4 on M1 at left.

- M1

8×4

Select A

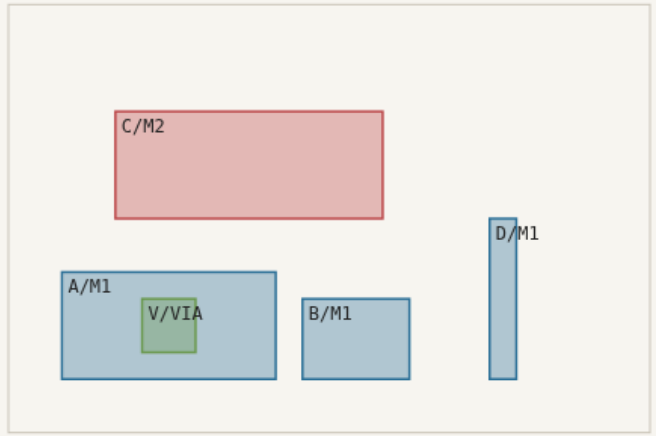
Stats

EDGES AND POLYGONS

Stats

Step 5 / 5 · stats · module01-01-edges-polygons

{"shapeCount":5,"layerCount":3,"layers":["M1","M2","VIA"],"edgeCount"



Explanation

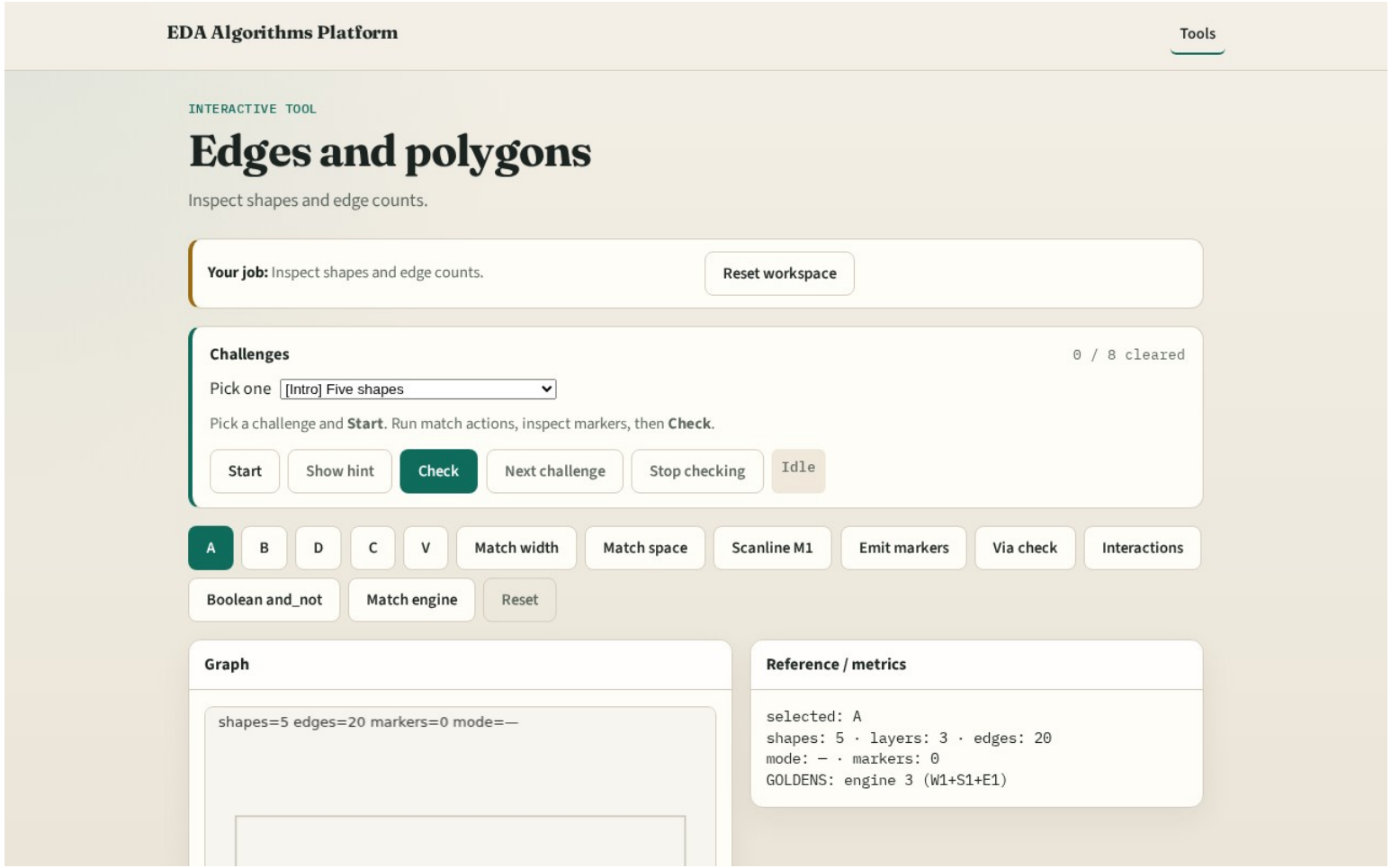
geometry_stats reports counts and widths.

- stats

```
GOLDEN 5/20
```

Stats

Browser lab track



Browser lab starter

Implement track

- Call geometry_stats(); assert shape_count==5 and edge_count==20

Pitfalls

- Counting only cells
- Mixing chip size with shape count

Your turn

- Next: single-layer edge-pair spacing